Chapter 13, Problem 94CP

Problem

A 440/110-V ideal transformer can be connected to become a 550/440-V ideal autotransformer. There are four possible connections, two of which are wrong. Find the output voltage of:

(a) a wrong connection,

(b) the right connection.

Step-by-step solution

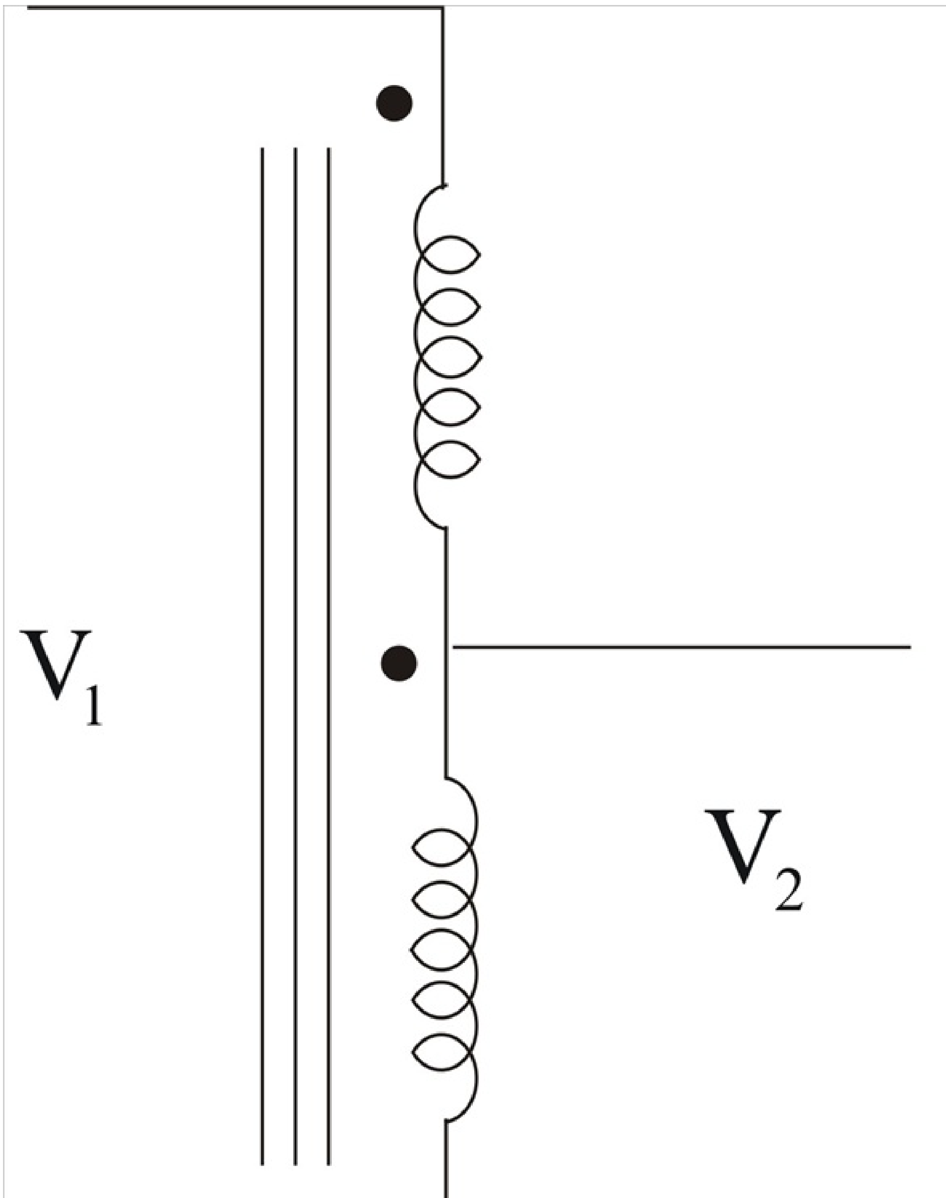
Step 1 of 5

$$\frac{V_2}{V_1} = \frac{110}{4}$$

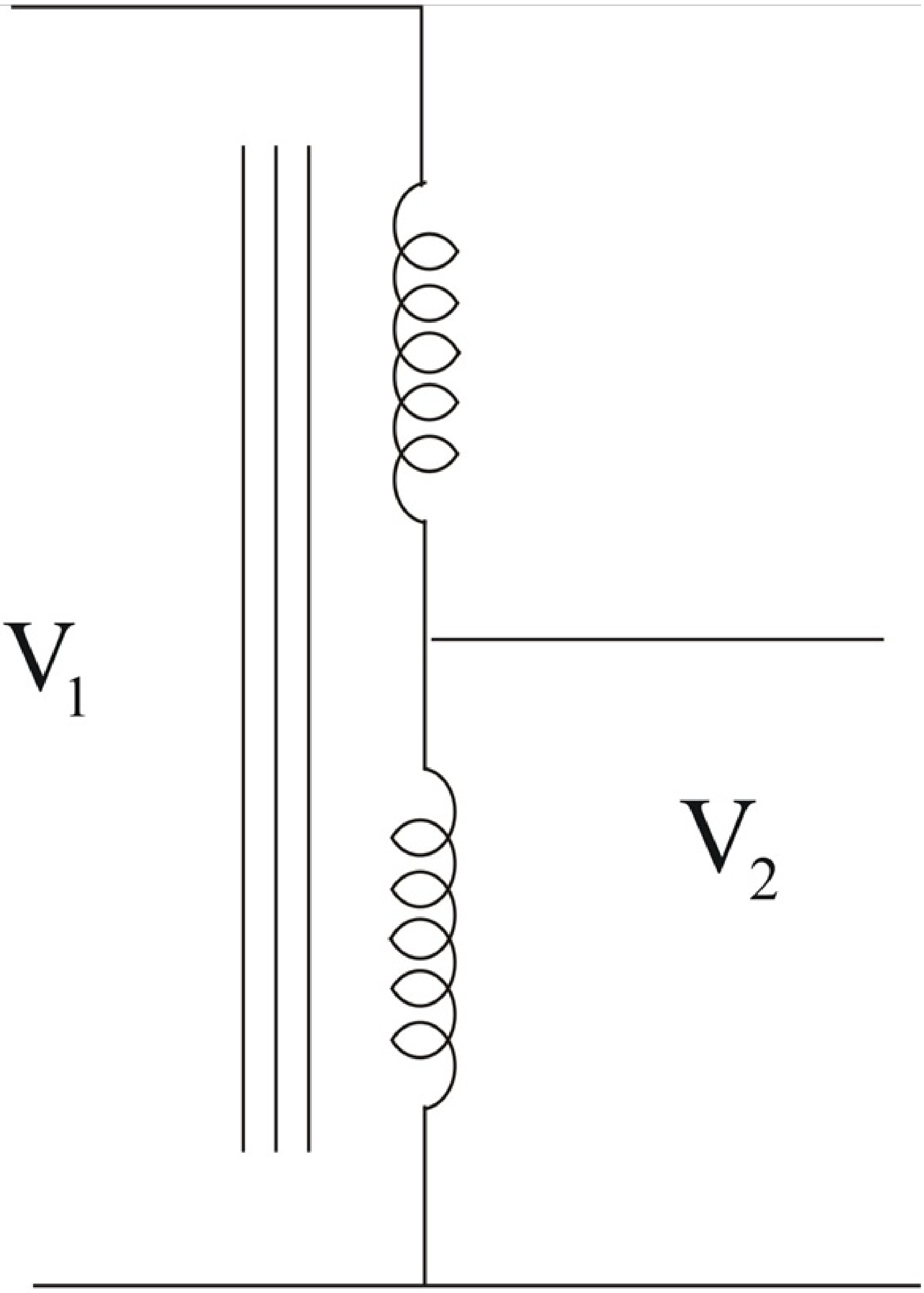
$$= \frac{1}{4} = \frac{I_1}{I_2}$$

There are four ways of hooking up the transformer as an auto – transformer; however it is clear that there are only two outcomes.

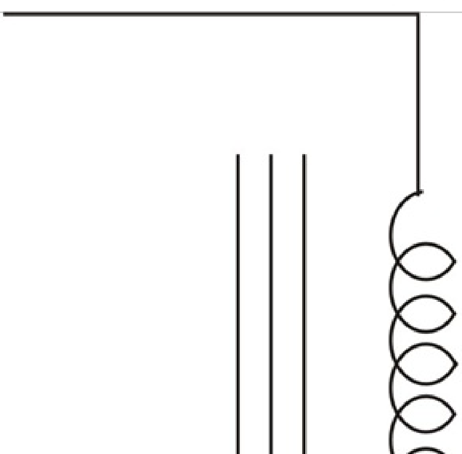
Step 2 of 5

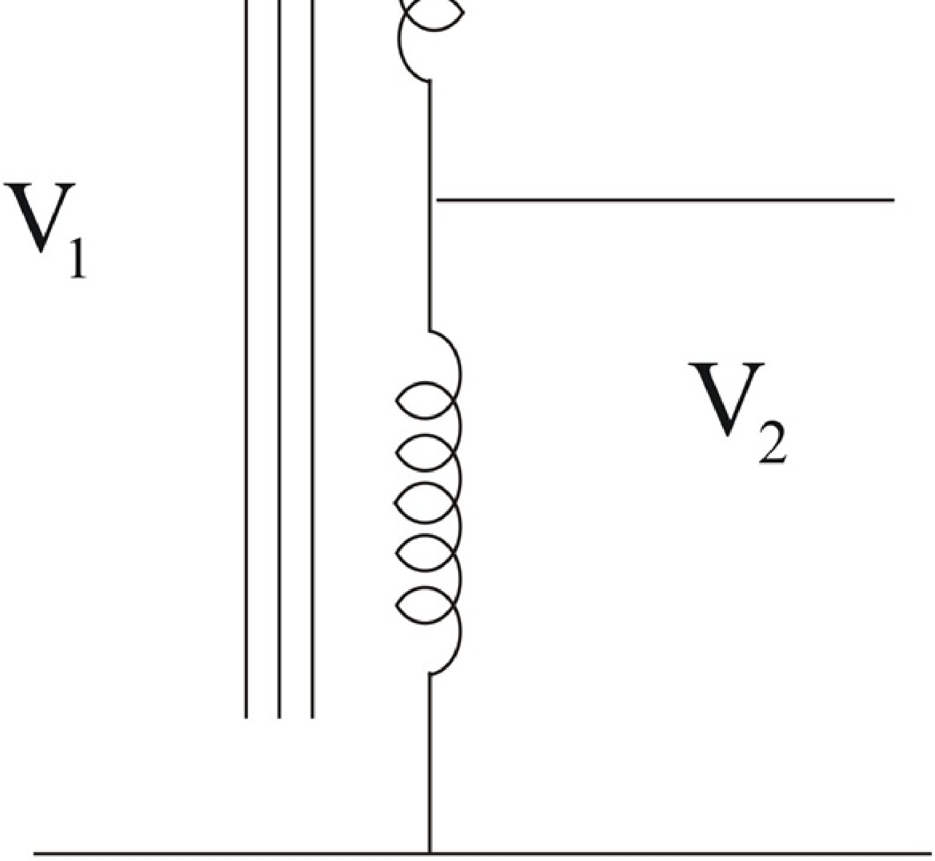


(1)

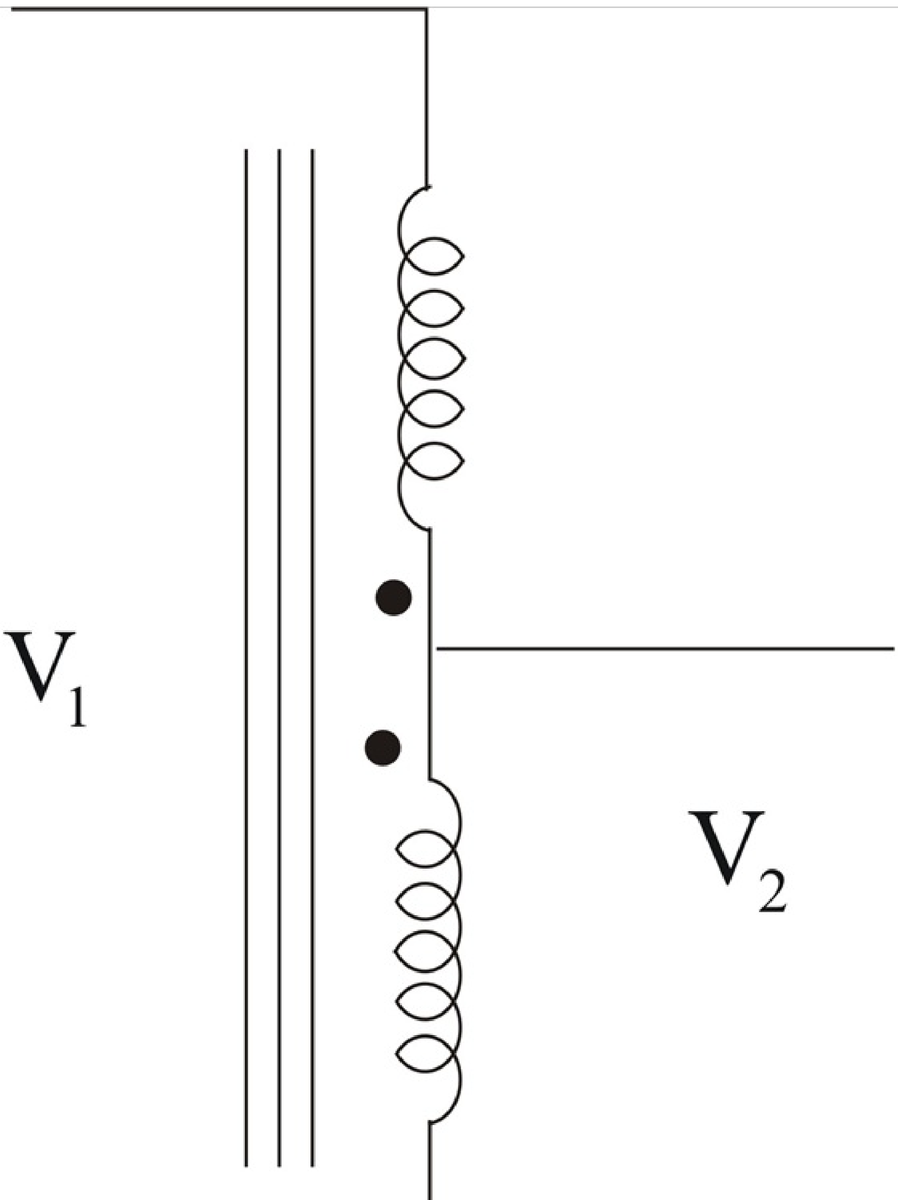


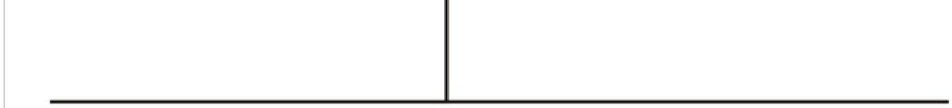
(2)





(3)





(4)

Step 3 of 5

(1) and (2) produce the same results and (3) and (4) also produce the same results. Therefore, we will only consider figure (1) and (3).

Step 4 of 5

(A) For figure (3),

$$\frac{V_1}{V_2} = \frac{550}{V_2}$$

$$= \frac{(440-110)}{440} = \frac{330}{440}$$

Thus,

$$V_2 = 550 \times \frac{440}{330}$$

$$= 733.4\ V_X\ \ [\,Not\ \ desired\ \ result\,]\ .$$

Step 5 of 5

(B) For figure (1),

$$\frac{V_1}{V_2} = \frac{550}{V_2}$$

$$= \frac{(440+110)}{440} = \frac{550}{440}$$

Thus,

$$V_2 = 550 \times \frac{440}{550}$$

$$= 440\ V\ \ [\,The\ \ desired\ \ result\,]\ .$$